



Getting Started with GitHub

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Latest Revision: 05/01/2025



What is GitHub?

- A collaborative platform for software development and project management
- Hosts repositories (repos) to store, track, and share project files and code
- Cloud based storage for all file types
- Based on the “Git” system

What is the purpose of GitHub?

- Version control: Tracks changes and allows rollback to earlier versions
- Collaboration: Enables multiple people to work together on projects simultaneously
- Task management: Organizes project tasks using built-in project boards
- Documentation: Maintain sustainability and knowledge transfer by showing the paths and details of the design/programming process
- File Storage: Allows us to store files on the cloud for use in future years



Why is it important? What do we use it for?

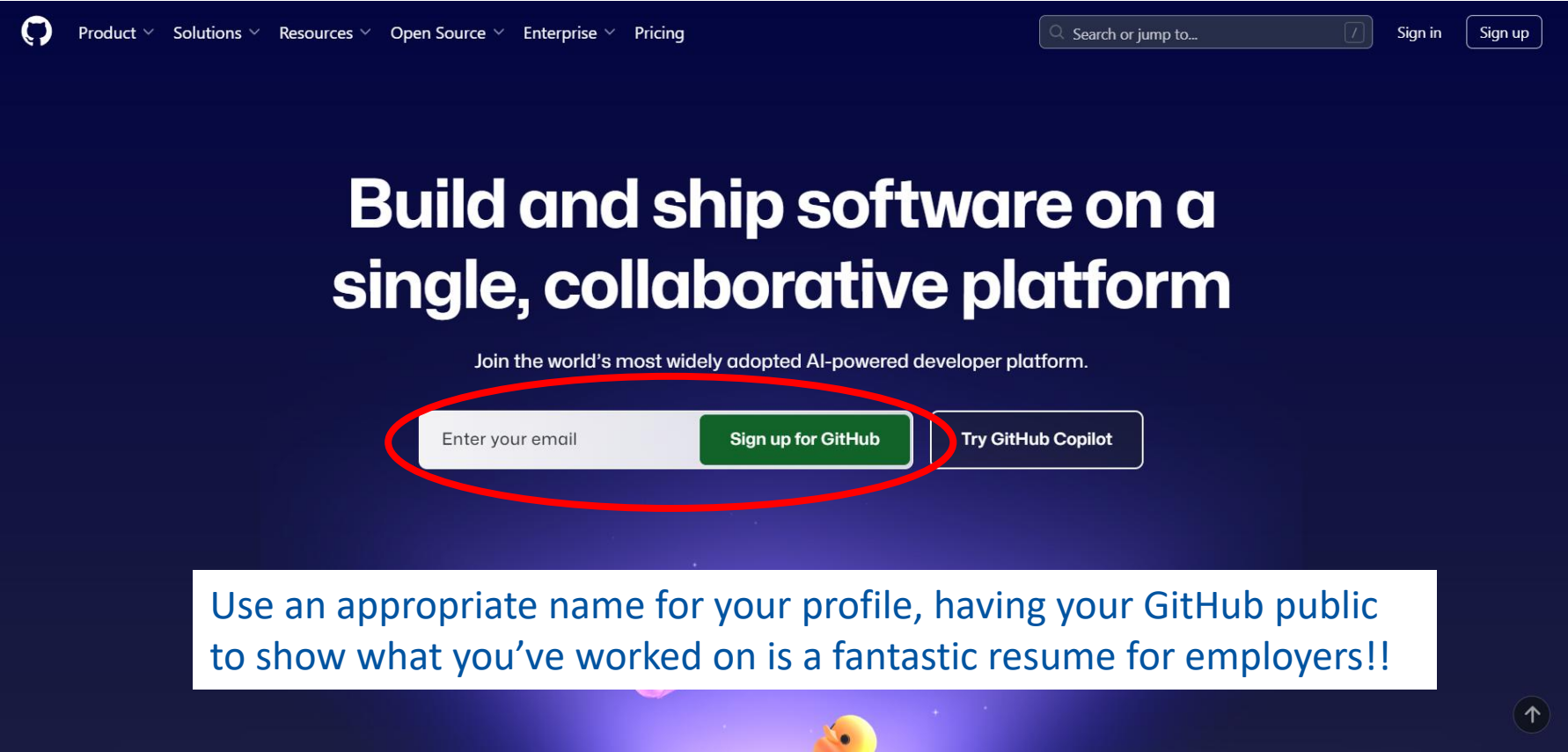
Programming:

- Stores the code for the robot
- Allows people to work on stuff at the same time
- Make modifications or test functionalities while maintaining the previous versions
- Organizes changes and notifies people of what has changed and where

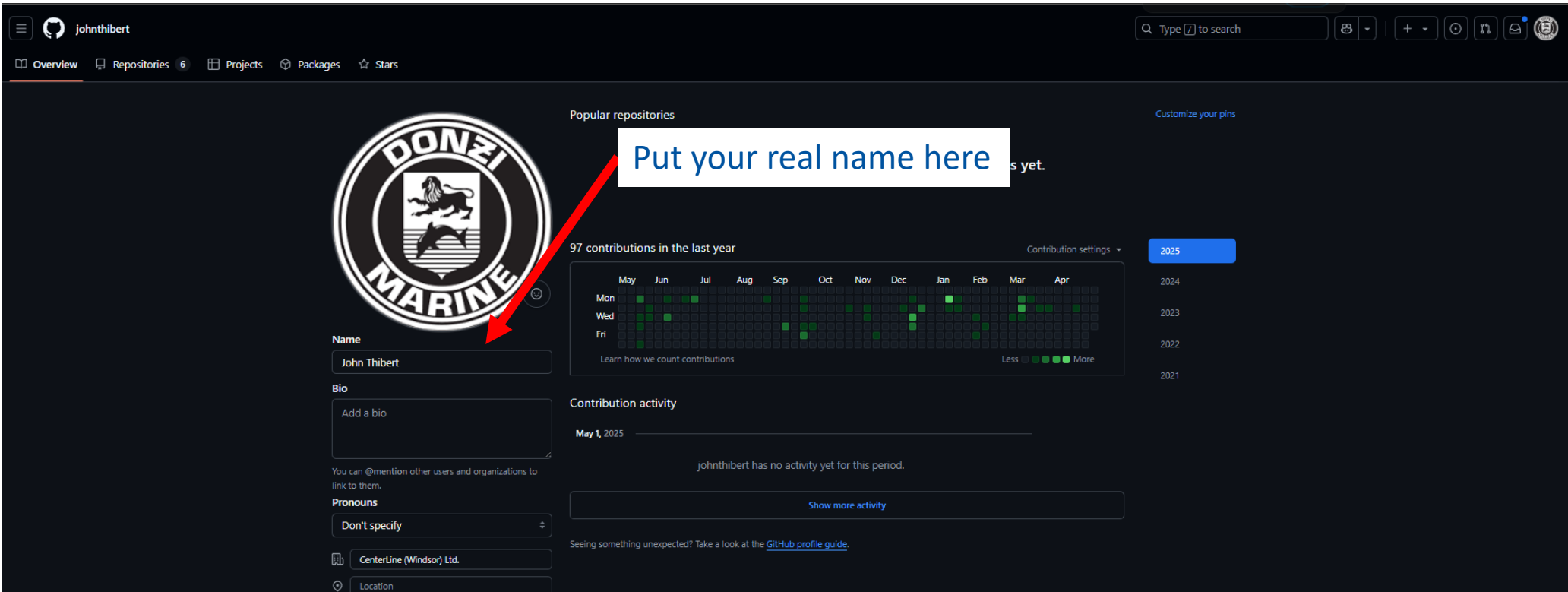
Mechanical:

- Stores robot CAD files
- Allows everyone to have access to the latest CAD
- Allows people to make changes and concepts to be reviewed before executing on the main robot
- Organizes tasks and issues associated with various subsystems
- Documents our design process for later use (judges and awards)

Creating a GitHub account:



Creating a GitHub account:



The screenshot shows the GitHub profile page for 'johnthibert'. A red arrow points from a text box that says 'Put your real name here' to the 'Name' input field, which currently contains 'John Thibert'. The profile also features a 'DONZI MARINE' avatar, a bio field with 'Add a bio', and a contribution graph showing 97 contributions in the last year. The 'Popular repositories' section is empty, and the 'Contribution activity' section shows no activity for the selected period (May 1, 2025).



Creating a GitHub account:

- Once you have created an account, contact a mentor to invite you to the FRC5885 GitHub page:

<https://github.com/frc5885>

- This will allow you to make changes and view archived and private repositories



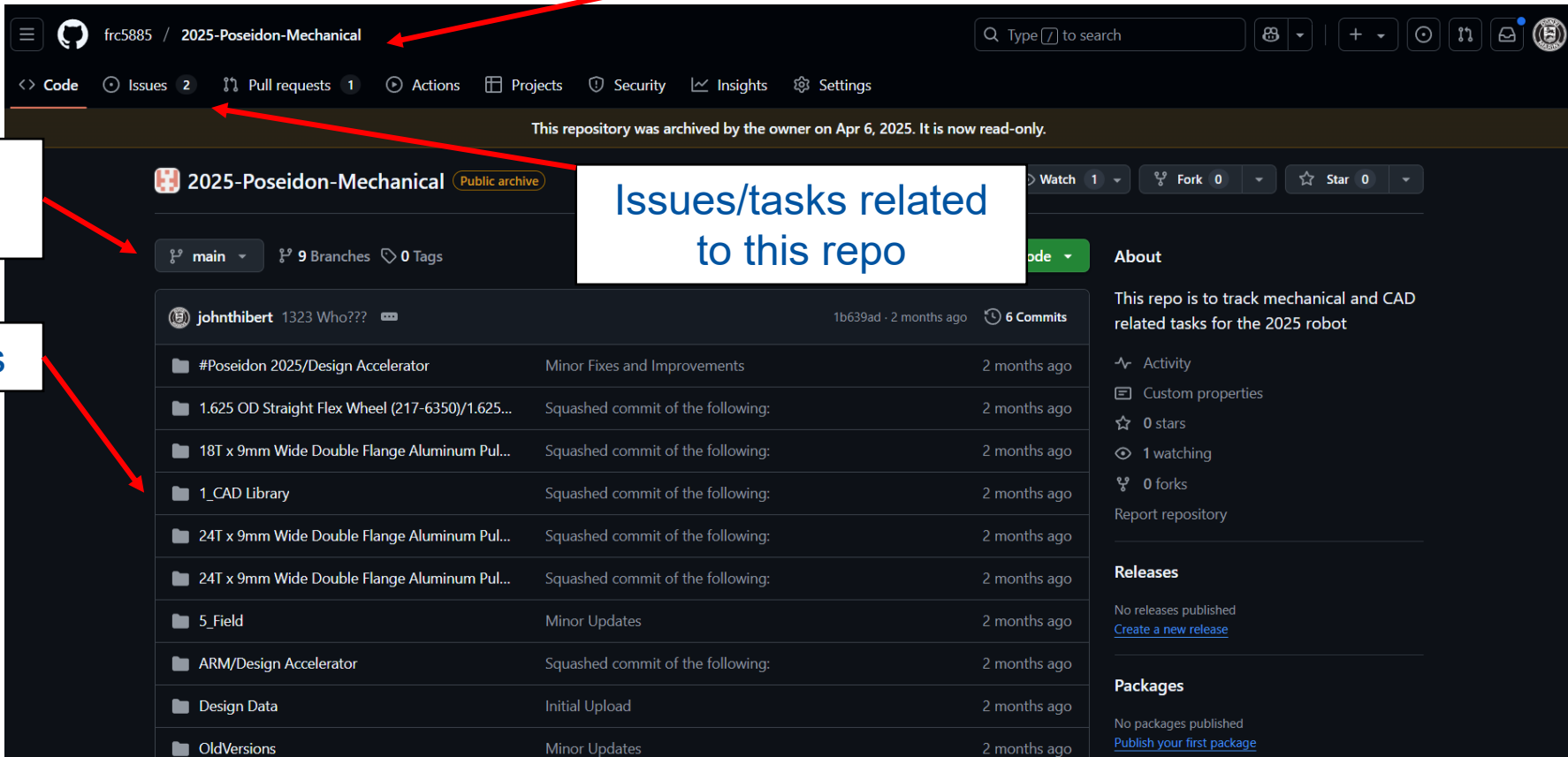
FRC5885 homepage

Repo's and projects can be found under these tabs of the FRC5885 homepage. Projects will be the most frequently accessed.



Intro to GitHub - Repositories

Poseidon mechanical repository



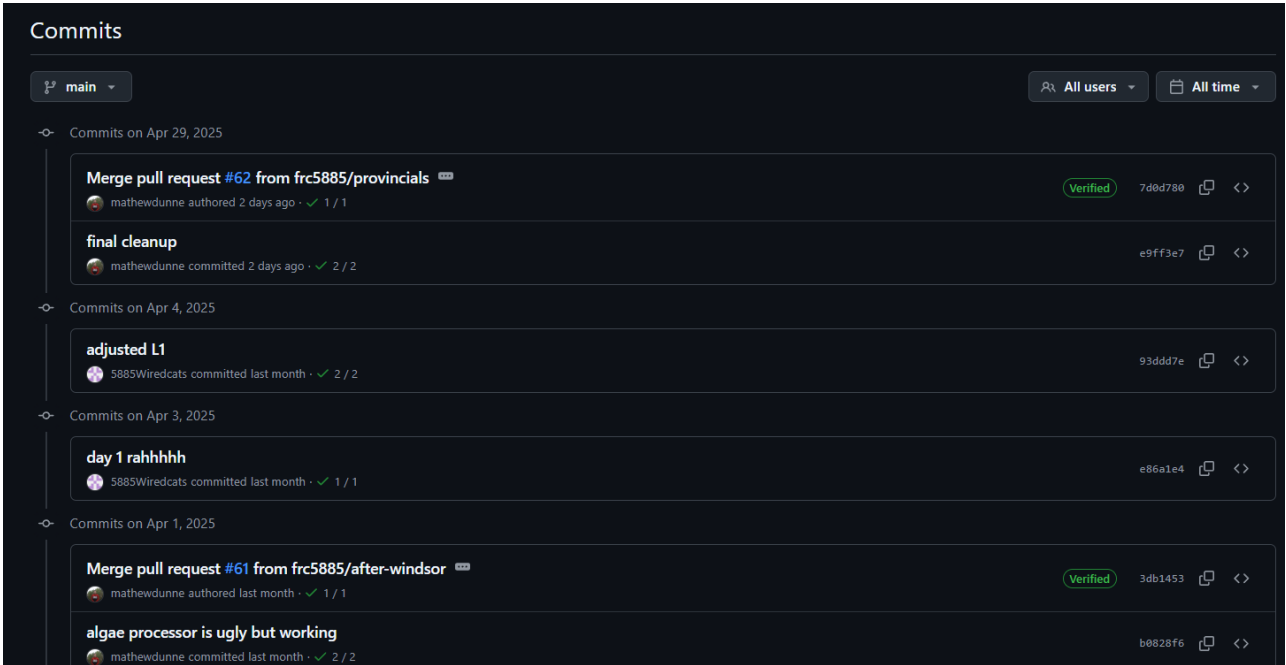
The screenshot shows the GitHub repository page for '2025-Poseidon-Mechanical' by user 'frc5885'. The repository is archived and read-only. The page includes a navigation bar with links to Code, Issues (2), Pull requests (1), Actions, Projects, Security, Insights, and Settings. A message states: 'This repository was archived by the owner on Apr 6, 2025. It is now read-only.' The repository name '2025-Poseidon-Mechanical' is displayed with a 'Public archive' badge. Below the name, it shows 'main' branch, '9 Branches', and '0 Tags'. The commit history table lists several commits by 'johnthibert' from 2 months ago, including files like '#Poseidon 2025/Design Accelerator', '1.625 OD Straight Flex Wheel', '18T x 9mm Wide Double Flange Aluminum Pul...', '1_CAD Library', '24T x 9mm Wide Double Flange Aluminum Pul...', '5_Field', 'ARM/Design Accelerator', 'Design Data', and 'OldVersions'. The right sidebar contains 'About' (This repo is to track mechanical and CAD related tasks for the 2025 robot), 'Activity', 'Custom properties', '0 stars', '1 watching', '0 forks', 'Releases' (No releases published), and 'Packages' (No packages published).

Annotations on the screenshot:

- Branch selection:** Points to the 'main' branch dropdown menu.
- CAD Files:** Points to the '1_CAD Library' folder in the commit history table.
- Issues/tasks related to this repo:** Points to the 'Issues' link in the navigation bar.

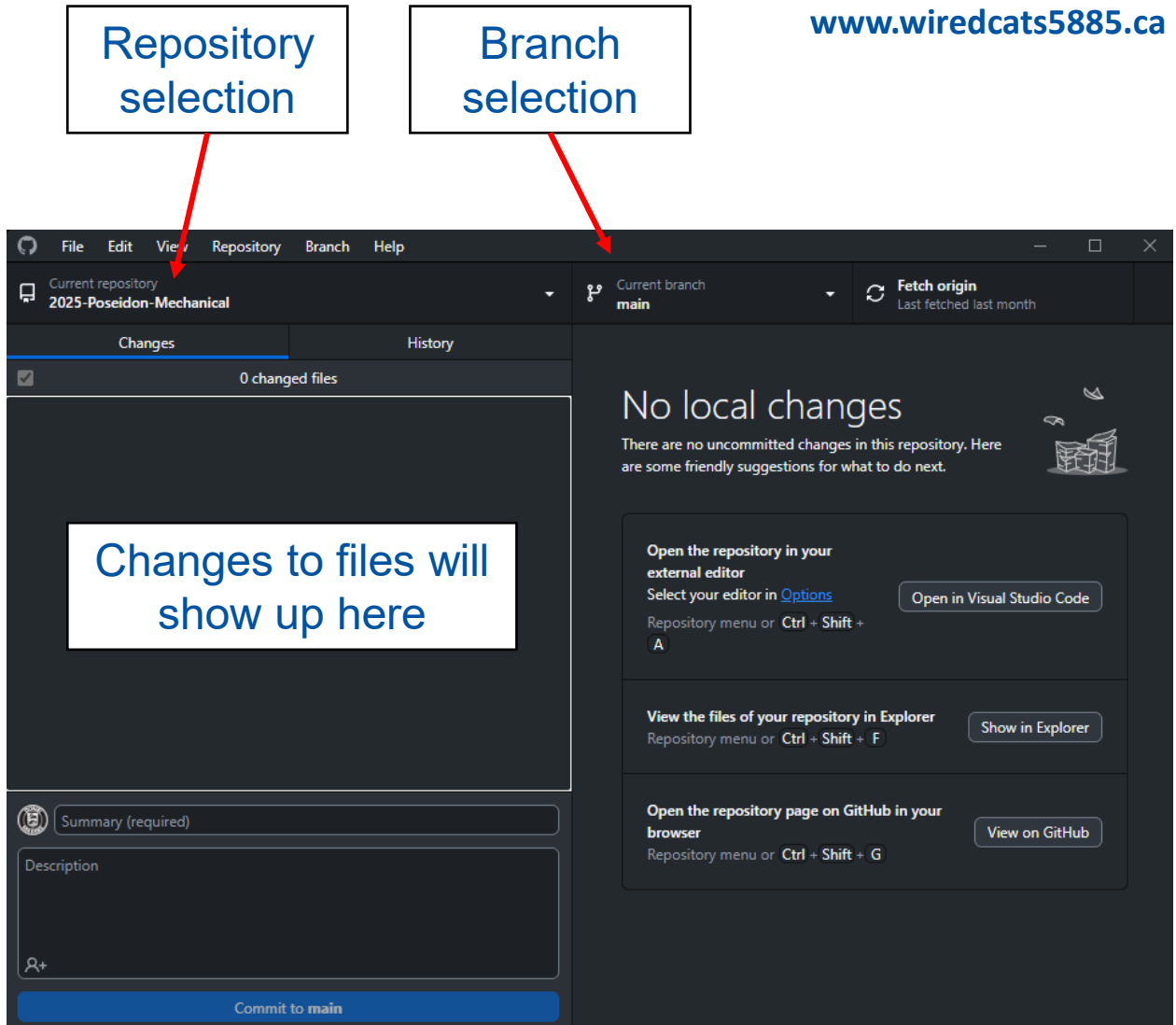
Intro to GitHub - Commits

- Commits are “check-ins”
- When you are working on a branch of code and want to save it to the repository (saving progress, completed a task, fixed an issue, added a feature) you use a commit to save that progress.
- Captures your work at that time



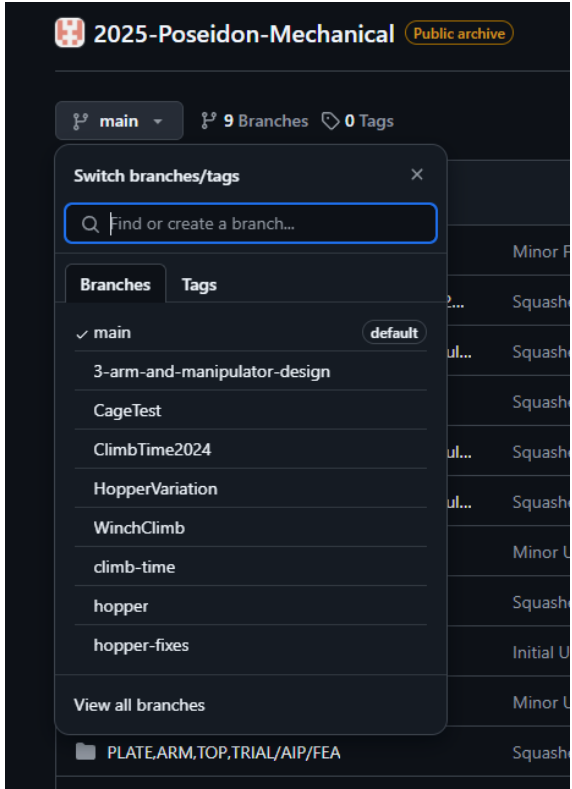
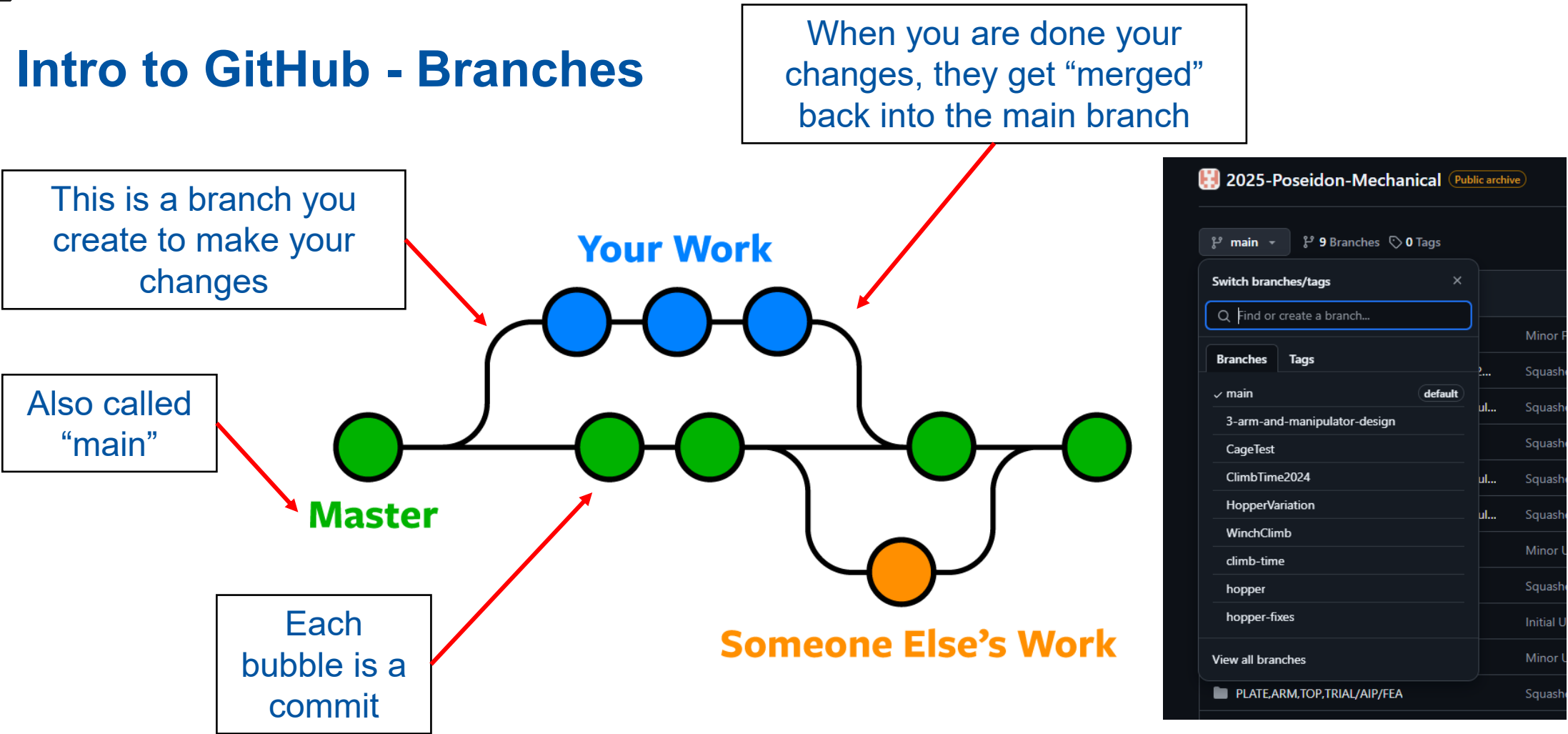
Intro to GitHub - Pushes

- Whenever a commit is made, that change is solely local.
- When you want to upload that commit to GitHub, you do so with a “push”.
- Pushes upload all your “stashed” commits to the branch you are working on



Shown above is the “GitHub Desktop” application interface – more on this later

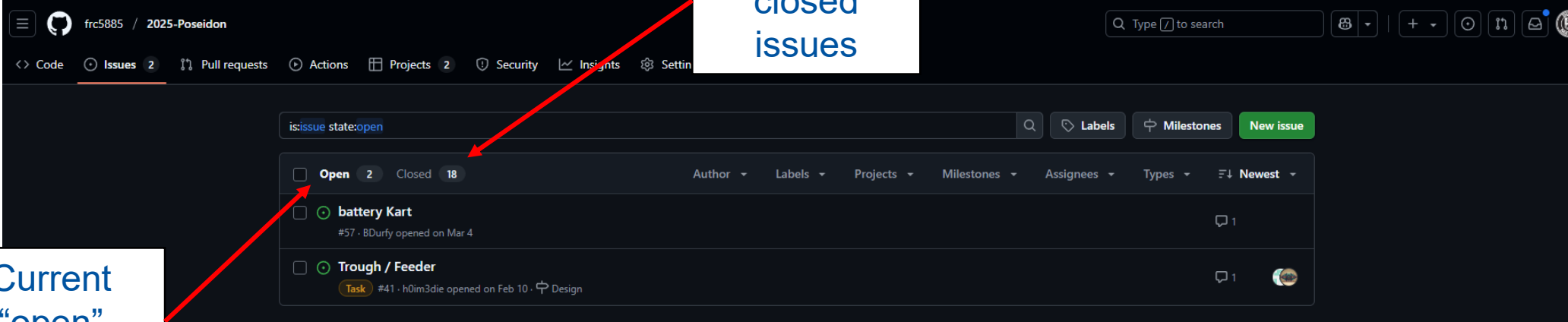
Intro to GitHub - Branches



Intro to GitHub - Issues

Current
“closed”
issues

Current
“open”
issues



The screenshot shows the GitHub interface for the repository 'frc5885 / 2025-Poseidon'. The 'Issues' tab is selected, showing a filter for 'Open' issues (2) and 'Closed' issues (18). Two red arrows point from the callout boxes to these filters. The list of issues includes 'battery Kart' and 'Trough / Feeder'.

Issue State	Count
Open	2
Closed	18

Issue Title	Author	Created
battery Kart	#57 · BDurfy	opened on Mar 4
Trough / Feeder	#41 · h0im3die	opened on Feb 10 · Design



Intro to GitHub - Issues

- Issues always need to be made to a repository
- Issues always need to be given a milestone and added to the project board

Tracks history of the issue and status

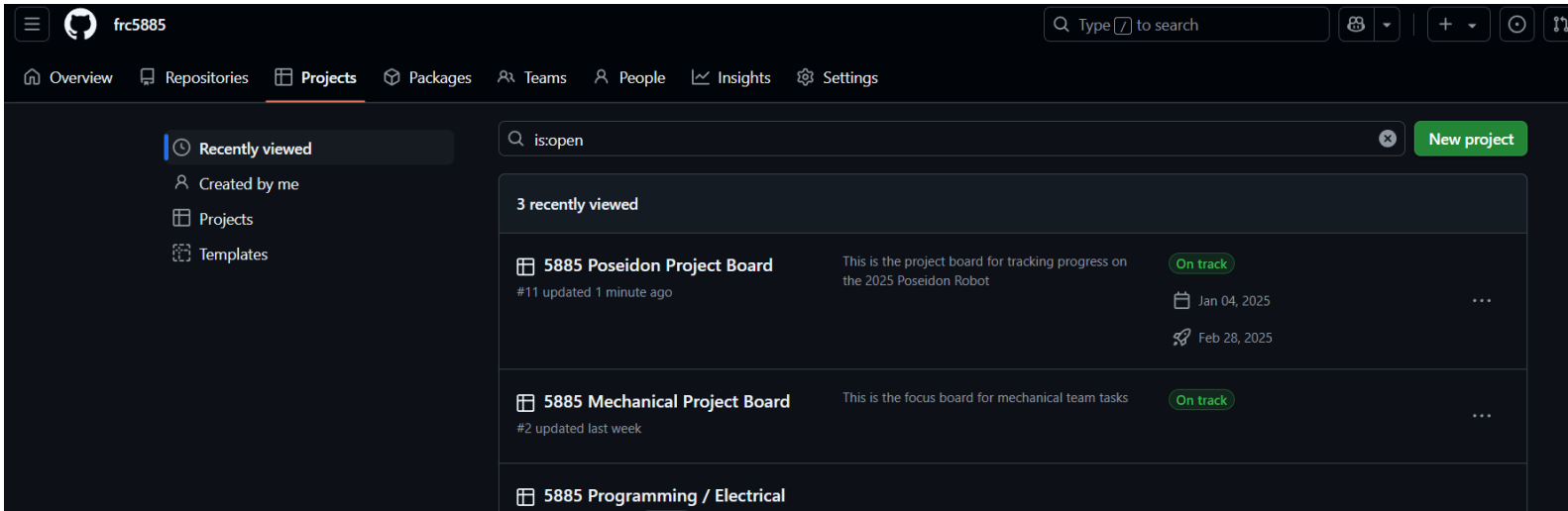
An update Owen provided when looking at 4946's feeder mechanism

Issue is currently an "open" issue

The screenshot shows a GitHub issue titled "Trough / Feeder #41" in a repository. The issue is marked as "Open" and "Task". It was opened by h0im3die on Feb 10. The description states: "No description provided." The issue has a history of updates, including assignments to marcotron64, zjs0221, and Vegancatburger, and links to the 5885 Poseidon Project Board and a Design milestone. A photo of a person working on a robot is included at the bottom. The right sidebar shows various settings and options for the issue.

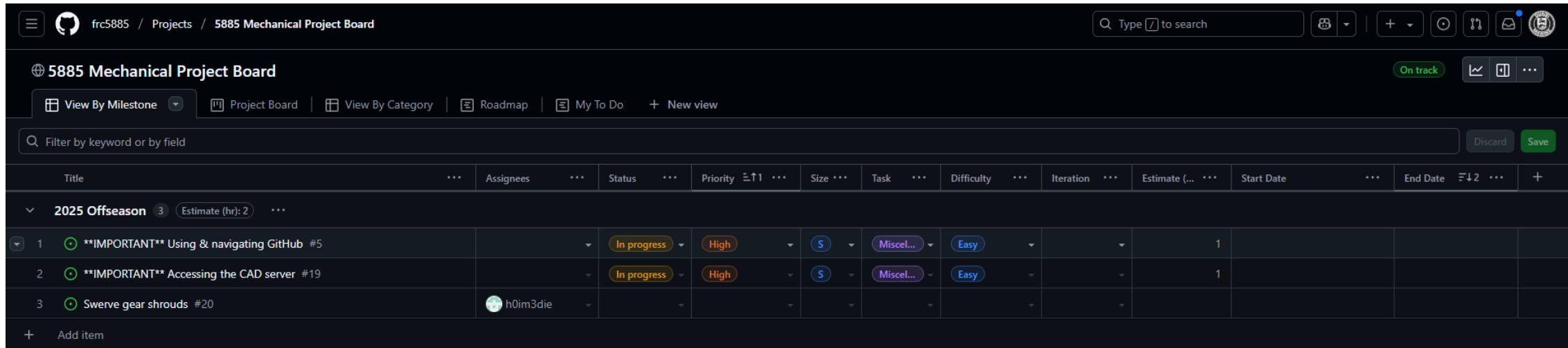
Intro to GitHub - Projects

- Projects allow us to organize issues, tasks, and deadlines all in one place
- Projects typically get created for large scale projects such as the 2025 robot



Intro to GitHub – Project Boards

- Project boards allow us to see all the current issues and tasks associated with that project
- The mechanical project board organizes this information to make sure important tasks are highly visible and tracks progress of them



The screenshot shows the GitHub Project Board for the 5885 Mechanical project. The board is titled "5885 Mechanical Project Board" and is currently set to "View By Milestone". The board is organized into columns representing different stages of the project: To Do, In Progress, and Done. The "In Progress" column contains three items:

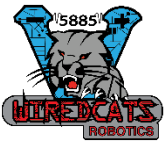
Title	Assignees	Status	Priority	Size	Task	Difficulty	Iteration	Estimate (hr)	Start Date	End Date
1. **IMPORTANT** Using & navigating GitHub #5		In progress	High	S	Miscel...	Easy		1		
2. **IMPORTANT** Accessing the CAD server #19		In progress	High	S	Miscel...	Easy		1		
3. Swerve gear shrouds #20	h0im3die									

At the bottom of the board, there is a button to "Add item".



See next presentation on GitHub workflow





Revision Table	
Date (MM/DD/YYYY)	Description
05/01/2025	First Release

